Chapter 21: Column, row, and null spaces

We're now going to connect the concept of dimension with the notion of rank.

In this section, we'll often realize vectors in $\mathbb{R}^n$ as rows instead of columns. The notions of span, linear independence, and basis are defined for rows.

Definition: Let

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

be an $m \times n$ matrix.

- The columns of A, considered as vectors in $\mathbb{R}^m$, span a subspace of $\mathbb{R}^m$ called the **column space** of A, denoted $\operatorname{col} A$.
- The rows of A, considered as vectors in $\mathbb{R}^n$ (viewed as rows instead of columns), span a subspace of $\mathbb{R}^n$ called the **row space** of A, denoted $\operatorname{row} A$.

Remark: The column space of A is exactly the same as the image of A.

Lemma: Let A and B be two $m \times n$ matrices.

a. If A and B are row equivalent matrices, then the row spaces of A and B are equal, i.e. $\operatorname{row} A = \operatorname{row} B$.

b. If A and B are column equivalent matrices, then the column spaces of A and B are equal, i.e. $\operatorname{col} A = \operatorname{col} B$.

Note: "Column equivalent" is the analogous definition of "row equivalent" when doing column operations instead of row operations.
Key idea in proof of (a):

Lemma: If R is a row-echelon matrix, then

1. The nonzero rows of R are a basis of $\operatorname{row} R$.
2. The columns of R containing leading ones are a basis of $\operatorname{col} R$.

Proof:

Remark: In Chapter 2, we defined the **rank** of a matrix A to be the number of leading 1s (i.e. the number of pivots) in any row-echelon form matrix R that is row-equivalent to A. Since $\operatorname{rank}(A) = \dim(\operatorname{row} A)$, this justifies that the rank of A is independent of choice of R.

Example: Find a basis for $V = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 2 & 1 & 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 & 2 & -3 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 & 1 & -2 \end{bmatrix}, \text{ and } \mathbf{v}_4 = \begin{bmatrix} 1 & 1 & -1 \end{bmatrix}.$$

Theorem: Rank Theorem
Let A denote any $m \times n$ matrix of rank r. Then

$$\dim(\operatorname{col} A) = \dim(\operatorname{row} A) =$$

Moreover, if A is carried to a row-echelon matrix R by row operations, then

1. The r nonzero rows of R are a basis of $\operatorname{row} A$.
2. If the leading 1s lie in columns $j_1, j_2, ..., j_r$ of R, then columns $j_1, j_2, ..., j_r$ of A are a basis of $\operatorname{col} A$. (Pick up the pivots!)

Example: Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix in the previous example.

1. What is the $\dim \operatorname{row} A$?
2. Find a basis for the column space/image space of A.
3. What is the $\dim \operatorname{col} A$?
4. Find a basis for the nullspace of A.

Definition: The **nullity** of a matrix A is the $\dim(\operatorname{null} A)$.

Theorem: Let A denote an $m \times n$ matrix of rank r. Then

1. The $n-r$ basic solutions to the system $A\mathbf{x} = \mathbf{0}$ provided by the Gaussian algorithm are a basis of $\operatorname{null}(A)$, so $\dim(\operatorname{null} A) = n - r$.
2. $\dim \operatorname{im} A = r$.
3. $\dim(\operatorname{im} A) + \dim(\operatorname{null} A) = n$.
4. In other words, $\operatorname{rank}(A) + \operatorname{nullity}(A) = n$

Fact: The following results follow from the Rank Theorem:
Let A be an $m \times n$ matrix.

a. $\operatorname{rank}(A) = \operatorname{rank}(A^T)$

b. If U is an $m \times m$ invertible matrix, and V is an $n \times n$ invertible matrix, $\operatorname{rank}(A) = \operatorname{rank}(UA) = \operatorname{rank}(AV)$.

Example: Given $B = \begin{bmatrix} 3 & -1 & 2 & 0 \\ -6 & 2 & -2 & 6 \end{bmatrix}$ and $RREF(B) = \begin{bmatrix} 1 & -1/3 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$, find a basis for $(\operatorname{im} B)$ and a basis for $\operatorname{null} B$.

Example: If U is the subspace of $\mathbb{R}^3$ given by $U = \operatorname{span}\left\{ \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}, \begin{bmatrix} -4 \\ 6 \\ 0 \end{bmatrix} \right\}$, find a basis for U.

Let's now add to our list of invertibility criteria:

Box Of Facts/Invertibility Criteria
Let A be an $n \times n$ matrix. The following are equivalent:

1. A is nonsingular/invertible.
2. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
3. A is row equivalent to I_n.
4. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.
5. $\text{rank}(A) = n$.
6. A can be written as a product of elementary matrices.
7. $\det(A) \neq 0$.
8. The null space of A consists of only the zero vector.
9. The columns of A are linearly independent.
10. The columns of A form a basis for $\mathbb{R}^n$.
11. The rows of A are linearly independent.
12. The rows of A form a basis for $\mathbb{R}^n$.

Chapter 22: Change of basis

Recall: If V be a vector space of dimension n, with ordered basis $B = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n)$ for V, then we can build a coordinate representation of $\mathbf{v}$:

$$C_B(\mathbf{v}) = C_B(a_1\mathbf{b}_1 + a_2\mathbf{b}_2 + \ldots + a_n\mathbf{b}_n) = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \cdots + a_n\mathbf{e}_n = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

Example: Consider P_1 with basis $B = (1, x)$. Find $C_B(p)$ for $p = x - 10$.

Consider P_1 with basis $E = (2x+1, 3x-2)$. Find $C_E(p)$ for $p = x - 10$.

What if there was a way to easily move from one basis to another?

Question: Suppose V is 3-dimensional, with $B = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ an ordered basis for V.

Suppose $\mathbf{v}$ and $\mathbf{w}$ are two vectors in V with $C_B(\mathbf{v}) = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$ and $C_B(\mathbf{w}) = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

What is $C_B(\mathbf{v} + \mathbf{w})$?

What is $C_B(k\mathbf{v})$ for a scalar k?

Fact: For a vector space V with ordered basis $B = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$,

- $C_B(\mathbf{v} + \mathbf{w}) = C_B(\mathbf{v}) + C_B(\mathbf{w})$ for $\mathbf{v}, \mathbf{w}$ in V
- $C_B(k\mathbf{v}) = kC_B(\mathbf{v})$ for $\mathbf{v}$ in V and k in $\mathbb{R}$.

Definition: Let $B = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ and $D = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$ be two ordered bases of an n-dimensional real vector space V.

Suppose $\mathbf{v}$ is a vector in V. We could use either basis B or D to write out a coordinate representation for $\mathbf{v}$:

$C_B(\mathbf{v}) = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$ means $\mathbf{v} =$

and $C_D(\mathbf{v}) = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$ means $\mathbf{v} =$

Goal of this section: Build a matrix P so that $PC_B(\mathbf{v}) = C_D(\mathbf{v})$ for all $\mathbf{v}$.

Let's view P as $P = [\mathbf{p}_1\ \mathbf{p}_2 \cdots \mathbf{p}_n]$ (i.e. the ith column of P is $\mathbf{p}_i$).

What is $C_B(\mathbf{v}_1)$?

If we want $PC_B(\mathbf{v}_1) = C_D(\mathbf{v}_1)$, what must be true about P?

More generally, if $PC_B(\mathbf{v}_i) = C_D(\mathbf{v}_i)$, what must be true?

P is called the **transition matrix/change matrix** from B to D, and is denoted $P_{D\leftarrow B}$.

Theorem: Let $B = (\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_n)$ and D denote ordered bases of a vector space V. Then the change matrix $P_{D\leftarrow B}$ is given in terms of its columns by

$$P_{D\leftarrow B} = \begin{bmatrix} C_D(\mathbf{v}_1) & C_D(\mathbf{v}_2) & \cdots & C_D(\mathbf{v}_n) \end{bmatrix}$$

and has the property that $C_D(\mathbf{v}) = P_{D\leftarrow B} C_B(\mathbf{v})$ for all $\mathbf{v}$ in V.

Procedure to build transition matrix/change matrix To compute the transition matrix $P_{D\leftarrow B}$ from B to D, we calculate the coordinate vector of each vector in B relative to the basis D. Then use these coordinate vectors, in order, as columns in the matrix $P_{D\leftarrow B}$.

Example: Let $B = (1, x)$ and $D = (x+1, x-1)$ be two ordered bases for P_1, and build $P_{D\leftarrow B}$.

Theorem: Let B and D denote ordered bases of a vector space V. Then we have the following properties:

1. $P_{B\leftarrow B} =$______

2. $P_{D\leftarrow B}$ is invertible and $P^{-1}_{D\leftarrow B} =$ __________

3. If E is another ordered basis of V, $P_{E\leftarrow D} P_{D\leftarrow B} =$__________

Example: As in the previous example, let $B = (1, x)$ and $D = (x+1, x-1)$ be two ordered bases for P_1. Compute $C_D(\mathbf{v})$ for $\mathbf{v} = 5 - x$ both (a) directly using D and (b) using $P_{D \leftarrow B}$.

Example: Let $B = (1, x)$ and $E = (2x+1, 3x-2)$. Verify that $P_{E \leftarrow B} = \begin{bmatrix} 3/7 & 2/7 \\ -2/7 & 1/7 \end{bmatrix}$, and use this to verify your answer for $C_E(x - 10)$.

Using $D = (x+1, x-1)$, use your previous answers to compute $P_{E \leftarrow D}$.